DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT

VIRTUAL TRAFFIC CONTROLLER

TEAM: IDEALISTIC SPARK

TEAM MEMBERS:

- Shivani U R
- Sarthak Mishra
- Mukesh Seervi

TEAM: IDEALISTIC SPARK

OBJECTIVE & OVERVIEW

PROBLEMS WITH TRADITIONAL TRAFFIC SYSTEMS:

- Fixed-time traffic lights lead to inefficiencies.
- Lack of real-time adaptability causes congestion.
- High fuel consumption and increased emissions.
- No predictive capabilities to foresee jams.

OUR SOLUTION:

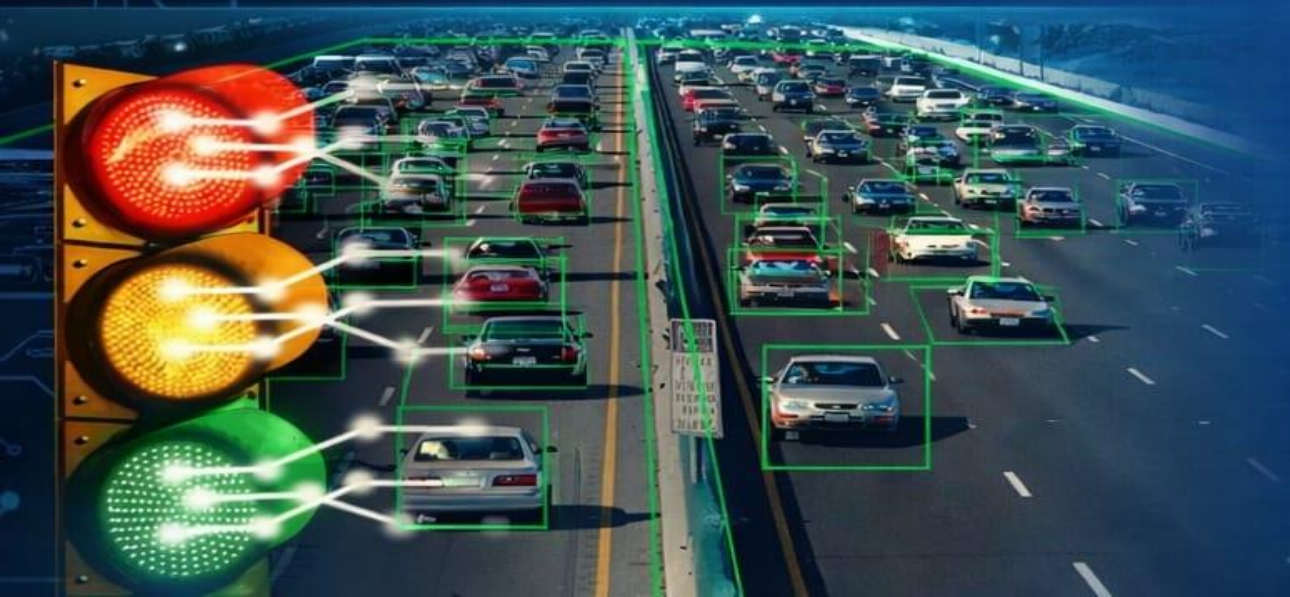
- ✓ Vision-based traffic controller using YOLOv8 & Python.
- ✓ Adaptive signal timing based on real-time congestion analytics.
- ✓ Predictive congestion algorithm forecasting jams in advance.
- ✓ Zero hardware beyond a laptop, ready for real-world deployment.

WORKING

VISION-BASED TRAFFIC CONTROLLER::

Components Used:

- Laptop + Webcam
- YOLOv8 (Real-Time Vehicle Detection)
- Python Script for Decision & Prediction



OUR WORKING PROJECT

System Architecture (Block Diagram)

1. Input Feed
Laptop Webcam or City CCTV Stream

→

2. Edge Vision
YOLOv8 AI detects & counts vehicles locally

3. Logic Engine
Python script calculates predictive congestion

→

4. Dashboard
Lovable/React UI displays live metrics

WORKING PIPELINE DETAILS

- Complete pipeline from camera to dashboard
  1. **Input Feed**: Laptop Webcam or City CCTV Stream
  2. **Edge Vision**: YOLOv8 AI detects & counts vehicles locally
  3. **Logic Engine**: Python script calculates predictive congestion
  4. **Dashboard**: Lovable/React UI displays live metrics
- **Data Flow:**
  - Video frames are processed strictly on the Edge device (Laptop). Only lightweight numerical data (vehicle counts, calculated wait times) is sent to the web dashboard via MQTT/WebSockets, ensuring low latency and high privacy.

PROTOTYPE ARCHITECTURE & SETUP

1. Simulated Traffic Feed

Running traffic footage on secondary screen, fed to Edge device.

Traffic Video Source

1. Simulated Traffic Feed

Running traffic footage on secondary screen, fed to Edge device.

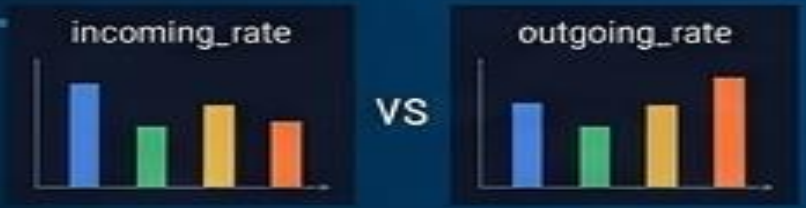
2. Edge Processing (Laptop):

Runs YOLOv8 and Logic Engine. Sends data to Web UI.

PROACTIVE TRAFFIC MANAGEMENT: The Algorithm

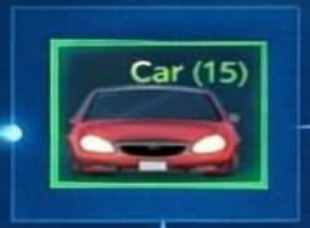
Moving from Reactive to Proactive

- System calculates Rate of Accumulation (RoA) over a rolling 60-second window.
- Identifies potential congestion BEFORE it occurs.
- Uses incoming_rate vs outgoing_rate comparison.



```
def adjust_green_light(duration):
    print(f"Green light duration set
to {duration} seconds")

def maintain_standard_cycle():
    print("Maintaining standard signal
cycle (30 seconds)")

def get_congestion_risk(queue_size):
    if queue_size > 80:
        return "CRITICAL"
    elif queue_size > 40:
        return "MODERATE"
    else:
        return "NORMAL"

def traffic_control(queue_size):
    congestion_risk =
get_congestion_risk(queue_size)

    if congestion_risk == "CRITICAL":
        predictive_delay = queue_size
* 2.5
        adjust_green_light(60)
    elif congestion_risk ==
"MODERATE":
        adjust_green_light(45)
    else:
        maintain_standard_cycle()

traffic_control(100)
traffic_control(50)
traffic_control(20)
```

TECHNICAL ASPECTS



1. VEHICLE DETECTION

Use YOLOv8 to identify & count vehicles from live video.

2. TRAFFIC ANALYTICS

Calculate traffic density & flow trends over time.

3. SMART SIGNAL TIMING

Adjust green light duration based on real-time congestion.

4. PREDICTIVE MODEL

Predict congestion 5 minutes ahead using recent data.

ALL DECISIONS MADE IN REAL-TIME ON LAPTOP (NO HARDWARE)

UNIQUENESS & IMPACT

FEASIBILITY

- **Fully software-based:** No hardware installation or maintenance needed.
- **Cost-effective:** Utilizes existing laptop webcam for real-time analysis.
- **Highly adaptable:** Easily portable to different traffic scenarios with internet access.

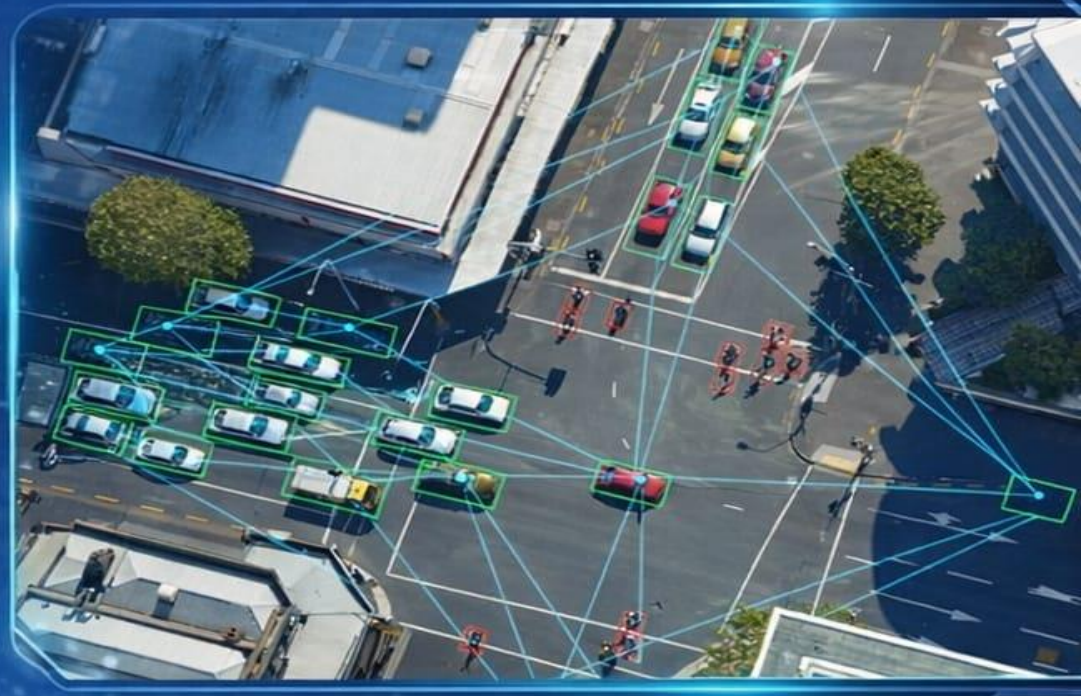
IMPACT

- **Reduced Congestion:** Intelligent signal control based on actual traffic conditions.
- **Eco-Friendly:** Lowers emissions by smoothing traffic flow, reduces fuel consumption.
- **City-wide Scalability.** Scalable solution for any urban or regional area.

ALL DECISIONS MADE IN REAL-TIME ON LAPTOP (NO HARDWARE)

CONCLUSION

- **Innovative gap** in traditional traffic control:
  Fills the void by predicting traffic jams before they occur.
- **Vision-based, hardware-free** solution:
  Efficient, adaptable, and cost-effective, using just a laptop and webcam.
- **Forecast-driven smart city tech**:
  Improves traffic flow and reduces emissions, paving the way for smarter urban areas.



THANK YOU!